Digital Forensics: CMP020N210S
Portfolio 1 – Analysing Forensic Images

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Introduction

In contemporary investigations, the field of digital forensics is essential since it allows analysts to retrieve and examine digital evidence. This portfolio offers practical forensic analysis expertise by examining forensic images and extracting possible evidence using industry-standard techniques

I operate as a digital forensics analyst on an incident response team, and one of my responsibilities is to look into a firm employee who may be involved in piracy and other illegal activities. USB_03.E01 and E_01_Physical_Image were two forensic images that were taken as digital evidence during the inquiry. My job is to look for concealed, erased, or questionable data in these pictures that might point to illegal activities.

I'll use forensic tools like FTK Imager, Autopsy, HxD, and VeraCrypt to undertake a comprehensive investigation, which will include tasks like file recovery, metadata analysis, encryption decryption, and locating hidden or erased data. Following the right forensic procedures will guarantee the accuracy and dependability of the results.

The five organised tasks in this portfolio—evidence extraction, data analysis, and forensic reporting—are all intended to improve my forensic investigation abilities. The results of the investigation will be useful in determining whether any criminal behaviour has occurred.

Investigator Details

Investigator detail	Date of the report	Purpose of the investigation
Safa Abiasis Bashir Yusuf	17/02/2025 – 10/03/2025	To Analyse forensic images for evidence

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Task 1) recovering files from a forensics Image

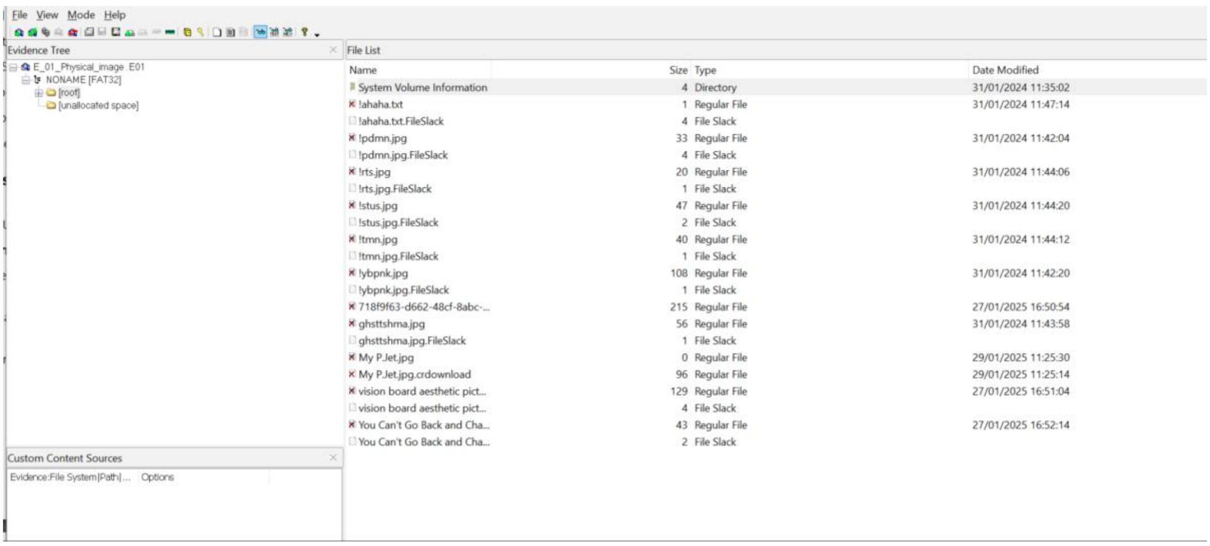
1.1 Identification of the person who added the files

Unfortunately, I was unable to determine who added the files to USB_Retro. I attempted a variety of techniques, including looking up timestamps, file information, and system statistics, but none of them provided a definitive response. No user information was present to associate the files with a particular individual.

1.2 File analysis: Names, Signatures, Count, Sizes and Extensions

I performed a forensic examination on the "Retro" USB drive and found that it had 21 files in total. Twelve of them are regular files, which are accessible and actively stored, and 9 of them are file slack entries, which are files that have residual data from earlier file deletions or changes.

- The screenshot below displays all 21 of the files—including ones with fax-related data—that are on the USB drive. Information that has been stored or maybe erased can be found using this file list. These files can be examined, their integrity checked, and pertinent data extracted for forensic analysis using FTK Imager.



- ***The 12 regular files are listed below, along with their names, signatures, and sizes in the table.***

No	File Name	File Signature	File extension	File Size (Bytes)
1	!ahaha.txt	.txt	N/A (Plain Text File, No Header)	1
2	!pdmn.jpg	JPEG (FF D8 FF E0)	.JPG	33

3	!rts.jpg	JPEG (FF D8 FF E0)	.JPG	20
4	!stus.jpg	JPEG (FF D8 FF E0)	.JPG	47
5	!tmn.jpg	JPEG (FF D8 FF E0)	.JPG	40
6	!ybpnk.jpg	JPEG (FF D8 FF E0)	.JPG	108
7	718f9f63-d662-48cf-8abc-4e29e428a5a1	JPEG (FF D8 FF E0)	.JPG	215
8	ghtsthma.jpg	JPEG (FF D8 FF E0)	.JPG	56
9	My P.Jet.jpg	JPEG (FF D8 FF E0)	.JPG	0
10	My P.Jet.jpg.crdownload	Chrome Download (Partial File)	.crdownload	96
11	Vision board aesthetic picture	JPEG (FF D8 FF E0)	.JPG	120
12	You Can't Go Back and Change the Beginning, Printable Wall Art, Inspirational Quote, Quote Print, Gift, Family Gift ETSY UK	JPEG (FF D8 FF E0)	.JPG	43

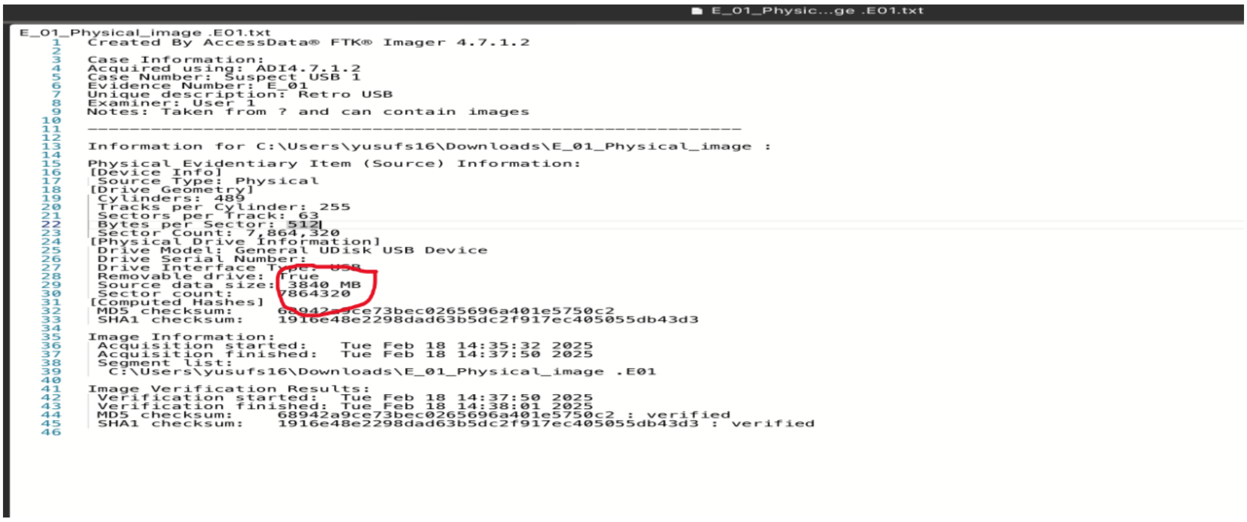
1.3 Storage Size of the USB Device

The USB device has a storage capacity of 3840 MB (3.84 GB). On ***line 29***, in the "[Physical Drive Information]" part of the document, this information appears clearly as follows: ***"Source data size: 3840 MB"***

This demonstrates that the "General UDisk USB Device" USB device has a 3.84 GB total capacity. To ensure accuracy and data integrity, AccessData® FTK® Imager 4.7.1.2 was used for the capture.

Evidence:

This information is visually confirmed in the screenshot below:



```
E_01_Physical_image .E01.txt
Created By AccessData® FTK® Imager 4.7.1.2

Case Information:
Acquired using: ADI4.7.1.2
Case Number: Suspect USB 1
Evidence Number: E_01
Unique description: Retro USB
Examiner: User 1
Notes: Taken from ? and can contain images

--------------------------------------------------------------

Information for C:\Users\yusufs16\Downloads\E_01_Physical_image :

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 489
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 7,864,320
[Physical Drive Information]
 Drive Model: General UDisk USB Device
 Drive Serial Number:
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 3840 MB
 Sector count: 7864320
[Computed Hashes]
 MD5 checksum:    68942a9ce73bec0265696a401e5750c2
 SHA1 checksum:   1916e48e2298dad63b5dc2f917ec405055db43d3

Image Information:
 Acquisition started:   Tue Feb 18 14:35:32 2025
 Acquisition finished:  Tue Feb 18 14:37:50 2025
 Segment list:
  C:\Users\yusufs16\Downloads\E_01_Physical_image .E01

Image Verification Results:
 Verification started:  Tue Feb 18 14:37:50 2025
 Verification finished: Tue Feb 18 14:38:01 2025
 MD5 checksum:    68942a9ce73bec0265696a401e5750c2 : verified
 SHA1 checksum:   1916e48e2298dad63b5dc2f917ec405055db43d3 : verified
```

1.4 Image of the USB device used in the analysis

This forensic analysis's actual USB device is shown in the image below. In order to generate a forensic image and retrieve possible digital evidence, this USB was analysed using FTK Imager. File extraction, storage capacity verification, and a search for erased data were all done by analysing the device.

Task 2 File Signature Investigations – Analysing the USB_03_E01 image

2.1 Adding the Image to FTK Imager and Verifying the Hash

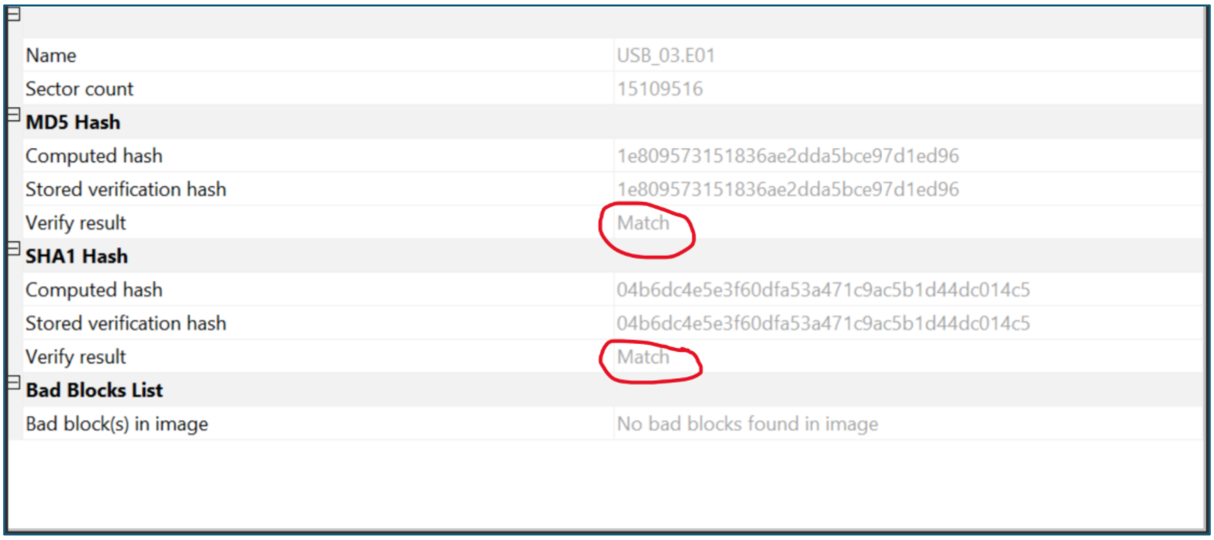
To add the downloaded image, I selected "File" > "Add Evidence Item" in FTK Imager, then selected "Image File" and loaded USB_03.E01.

I chose "Verify Drive/Image" when I right-clicked on the image in the Evidence Tree to confirm its integrity.

The image is a copy of the original with no changes, as confirmed by the computed MD5 and SHA-1 hashes matching the saved verification hashes.

Additionally, the **Bad Blocks List** showed no bad blocks, ensuring the image is intact and free from corruption.

As seen in the screenshot below, the verification results clearly indicate a **"Match"** for both hash values.



Name	USB_03.E01
Sector count	15109516
MD5 Hash	
Computed hash	1e809573151836ae2dda5bce97d1ed96
Stored verification hash	1e809573151836ae2dda5bce97d1ed96
Verify result	Match
SHA1 Hash	
Computed hash	04b6dc4e5e3f60dfa53a471c9ac5b1d44dc014c5
Stored verification hash	04b6dc4e5e3f60dfa53a471c9ac5b1d44dc014c5
Verify result	Match
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

2.2. Identifying File Formats Using HEX Signatures in FTK Imager

I found seven files with the names file01 through file07 in the root/Private folder of FTK Imager, which I used for this research. I extracted the file signatures using the Hexadecimal (HEX) View tab to ascertain their proper forms.

The table below records the file size, HEX signature, and corresponding file extension for each file.

File Name	File Size (Bytes)	File Signature (hex)	File extension
file 01	126 KB	24 50 44 46	.pdf
file 02	13,214 KB	50 4B 03 04	.zip

file 03	3 KB	none	.txt
file 04	79 KB	31 21 44 4F 43 54 59 50 45	.html or .htm
file 05	1,643 KB	37 7A BC AF 27 1C	.7z (7 -ZIP)
File 06	1,688 KB	47 49 46 38 39 61	.gif
file 07	53 KB	FF FB 90 44	.mp3

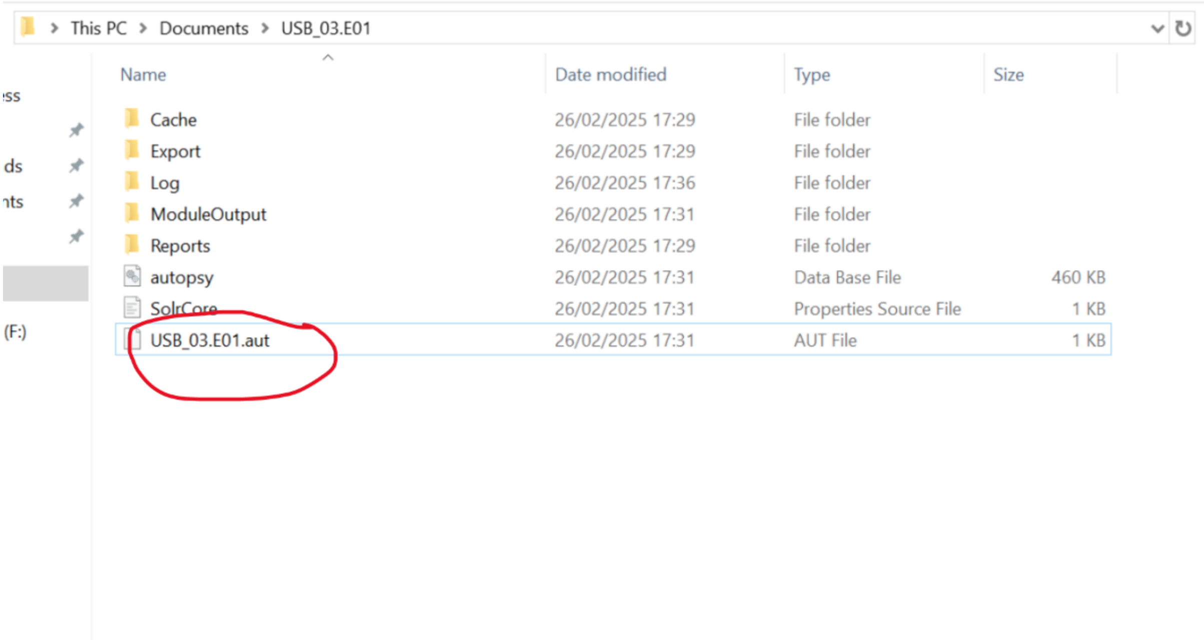
Task 3. Autopsy Case Analysis: USB_03.E01 Investigation

The investigation involves examining file metadata, identifying device information, analysing timeline events, and detecting potential encrypted files. The findings will be based on forensic analysis conducted within **Autopsy**; a digital forensics tool designed for in-depth file system examination.

3.1 Autopsy case file extension

After I created my Autopsy case, I went to the folder where I had saved it. I found a file inside the folder that had the extension .aut, confirming that Autopsy case files typically have this file extension.

The screenshot below shows that the case file is stored with the.aut extension, indicating that Autopsy saves forensic cases in this format.



3.2 Device Model Used to Create the **(Sword.Art.Online.full.1189342.jpg)**

Autopsy Device Identification

I opened Autopsy's Analysis Metadata and located the device model. The image Sword.Art.Online.full.1189342.jpg was created with an EPSON scanner, which is made by SEIKO EPSON CORP, according to the metadata.

Item: Sword.Art.Online.full.1189342.jpg
Aggregate Score: Not Notable

Analysis Result 1

Score: Not Notable
Type: EXIF Metadata
Configuration:
Conclusion:
Date Created: 2009-04-10 16:54:02 BST
Device Make: SEIKO EPSON CORP.
Device Model: EPSON scanner

Analysis Result 2

Score: Likely Notable
Type: Keyword Hits
Configuration:
Conclusion:
Keyword: Sword.Art.Online.full.1189342.jpg
Keyword Preview: «sword.art.online.full.1189342.jpg«
Keyword Search Type: 0

Analysis Result 3

Score: Likely Notable
Type: Keyword Hits
Configuration:
Conclusion:
Keyword: Sword.Art.Online.full.1189342.jpg
Keyword Preview: «sword.art.online.full.1189342.jpg«
Keyword Search Type: 0

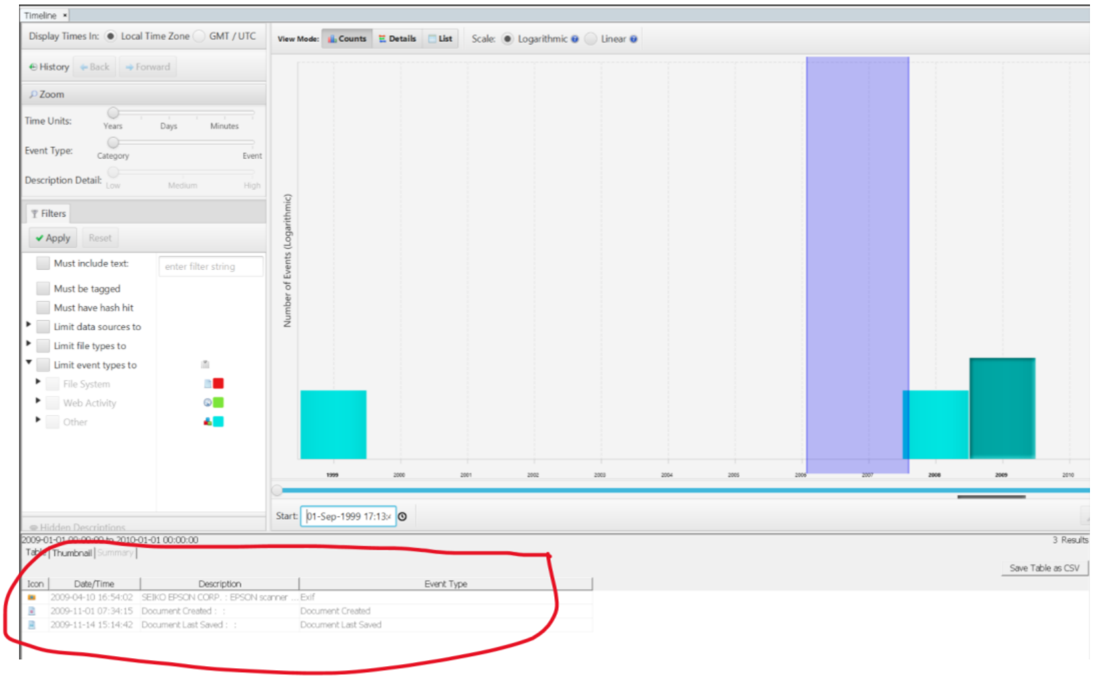
Analysis Result 4

Score: Unknown
Type: User Content Suspected
Configuration:
Conclusion:
Comment: EXIF metadata data exists for this file.

3.3 Types in Autopsy Timeline View for the Year 2009

I found three categories of events that took place in 2009 in the Autopsy Timeline view. These events consist of:

1. On April 10, 2009, Exif metadata was created and connected to an EPSON scanner.
2. On November 1, 2009, the document was created.
3. November 14, 2009, was the last time the document was saved.



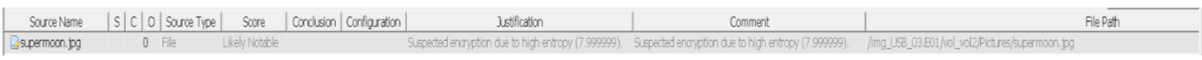
3.4 Locating a file that might be encrypted

According to the results of the forensic analysis of the USB_03.E01 picture using Autopsy, the following file was determined to be potentially encrypted because of its high entropy score.

- File Path: `supermoon.jpg`
- File Name: `/img_USB_03.E01/vol_vol2/Pictures/supermoon.jpg`
- Entropy Score: `7.999999 (indicating possible encryption)`

To conclude this task the file may be encrypted, heavily compressed, or concealing more data, according to the high entropy value. Forensic analysis is required to determine its true nature.

Below is a screenshot of the evidence.



Source Name	S	C	O	Source Type	Score	Conclusion	Configuration	Justification	Comment	File Path
supermoon.jpg			0	File	Likely Notable			Suspected encryption due to high entropy (7.999999).	Suspected encryption due to high entropy (7.999999).	/img_USB_03.E01/vol_vol2/Pictures/supermoon.jpg

Task 4: Analysing USB_03.E01 Forensic Image

In this task, we use Autopsy, FTK Imager, and VeraCrypt to analyse the USB_03.E01 forensic picture in order to find and unlock an encrypted file. Based on screenshots, we create a SHA256 hash to access the file's contents, list the files, and identify any suspicious or important information

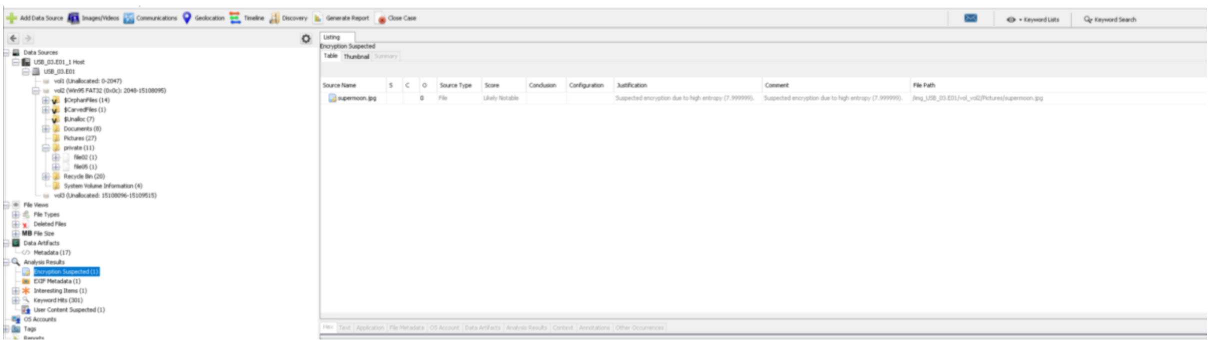
4.1 Locating the File to Open the VeraCrypt Container

The *"supermoon.jpg"* file has been marked as "Encryption Suspected" in Autopsy because of its high entropy (7.999999) based on Autopsy's forensic analysis of the USB_03.E01 picture. This shows that it might have a VeraCrypt container that is encrypted.

- **File Path: /img_USB_03.E01/vol_vol2/Pictures/supermoon.jpg**
- **File Name: supermoon.jpg**
- **Findings: High entropy in the file, which is showing us possible encryption.**

This file will be used to generate the SHA256 hash, extract and mount the VeraCrypt container, and examine its contents in the next steps.

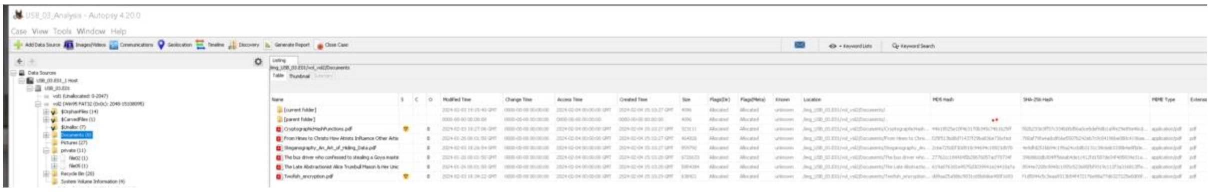
The screenshot is included below as proof



4.2 Finding the Right SHA256 to access the VeraCrypt Container

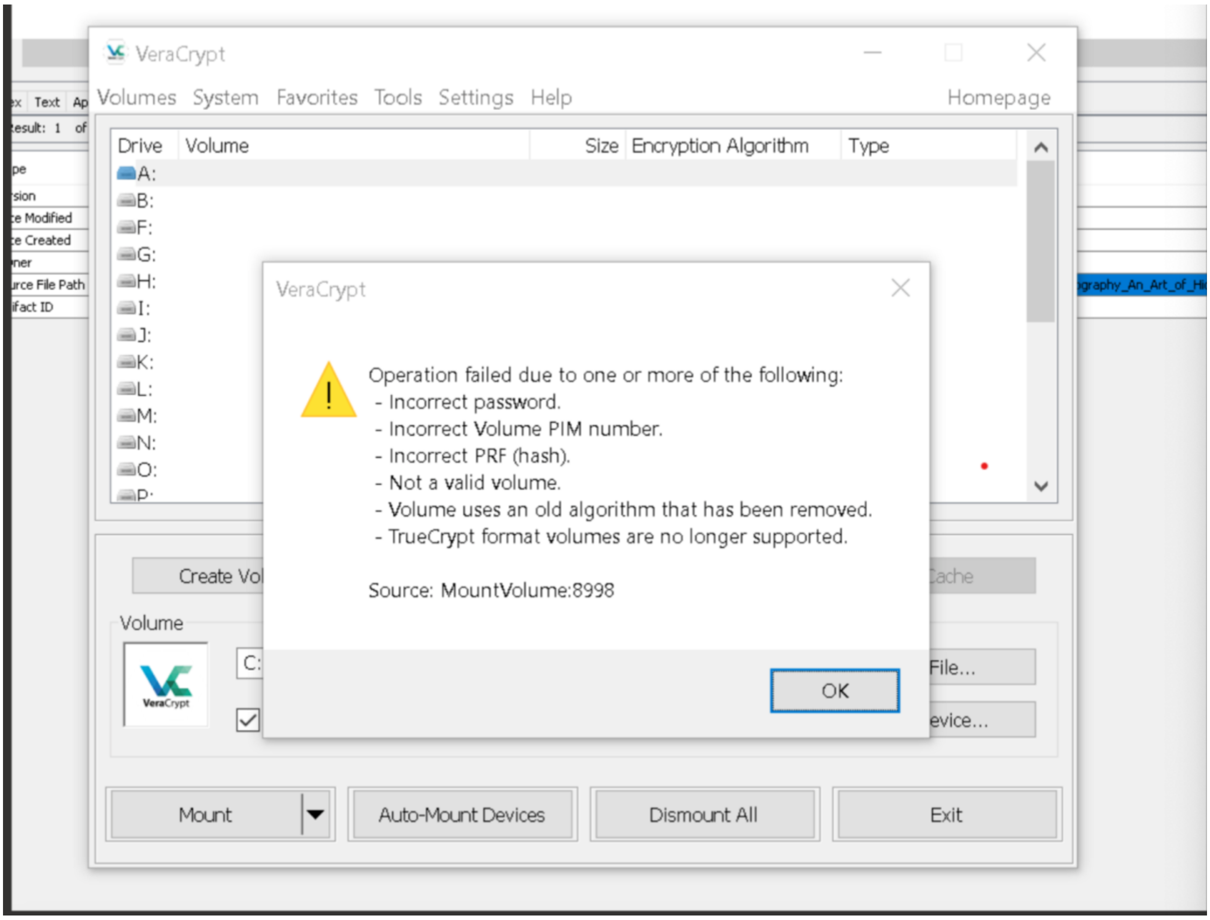
Screenshot 1:

The screenshot below displays the document files found on the USB in Autopsy. *Twofish_encryption.pdf* and *Steganography_An_Art_of_Hiding_Data.pdf* were among the six files that were recovered in total. For these files, the task was to create SHA256 hashes and use VeraCrypt to test them:



Screenshot 2:

The screenshot below is showing us an invalid format, PIM, or incorrect password prevented the container from being mounted, according to the VeraCrypt issue displayed in the picture. I fixed this by methodically checking the SHA256 hash of each file one after the other until I discovered the one that unlocked the container.



Screenshot 3:

In the terminal screenshot, you can see that two of the SHA256 pdf files was tested.

```
(base) safayusuf@safas-MacBook-Air ~ % echo -n "Steganography_An_Art_of_Hiding_D
ata.pdf" | shasum -a 256

86348a07558bce3208ab3bc1ab7f3fa03345aad6856a7e10bd9f7beae867adc6  -
(base) safayusuf@safas-MacBook-Air ~ % echo -n "Twofish_encryption.pdf" | shasum
 -a 256

cbca79cfd838aaae8b21f3df724fde090de17cb276dd82d78adfb48d984a36ae  -
(base) safayusuf@safas-MacBook-Air ~ %
```

Testing showed that the right "password" to open the VeraCrypt container is the SHA256 hash for Twofish_encryption.pdf.

ANSWER:

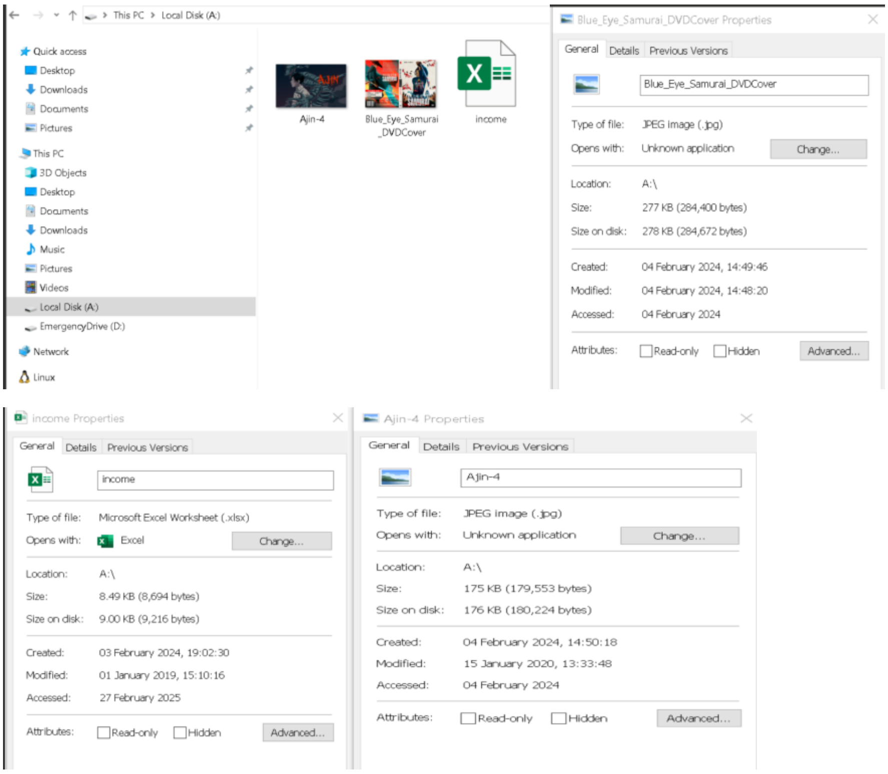
this sha256 was generated to access the VeraCrypt container through the twofish_encryption file.

cbca79cfd838aaae8b21f3df724fde090de17cb276dd82d78adfb48d984a36ae

4.3 Files in the VeraCrypt Container

I found three files on the Local Disk (A:), which looks to be a mounted VeraCrypt container: ***Ajin-4, Blue_Eye_Samurai_DvDCover, and Income***. These files were found by using File Explorer to navigate to Local Disk (A:), where they were clearly stored inside the container.

As an extra visual verification of the files found, a screenshot has been added. For reference, the file names, sizes, and extensions are listed in the data table after the screenshots:



Tables of files found:

File Name	File Size (Bytes)	File extension
Ajin-4	175 KB (179,553 bytes)	JPEG (.jpg
Blue_Eye_Samurai_DvDCover	277 KB (284,400 bytes)	JPEG (.jpg)
Income	8.49 KB (8,694 bytes).	.xlsx

4.4 Amount of Collectors found in the Income.xlsx file

21 collectors are listed in the "Collector" column of the Income.xlsx file, based on my review. Each name is associated with a person who has sold anime. As we can see, the table below clearly shows 21 individual entries, which is confirming the count. Visual proof of this data can be found in the screenshot below.

Final Answer:

- In total, there are 21 collectors in the income.xlsx file.

D26

	A	B	C	D	E
1	**Collector**	**Years as client**	**Number of anime sales**	**Average sale price**	**Total income**
2	Jerry Dyson	1	81	£12	£972.00
3	Nancy Logan	1	12	£21	£252.00
4	Chris Deyl	2	67	£11	£737.00
5	Jodi S. Boling	2	79	£9	£711.00
6	Ermias Adonay	1	11	£20	£220.00
7	Norm Rico	1	4	£30	£120.00
8	Haddas Michael	1	12	£25	£300.00
9	Al Taiqraa	1	1	£60	£60.00
10	Vin Alvaro	2	87	£9	£783.00
11	Faye Fraser	1	8	£25	£200.00
12	Divya Jatin	2	71	£20	£1,420.00
13	Hirato kausaki	1	2	£56	£112.00
14	Galra Etik	1	15	£25	£375.00
15	Riya Naveen	1	9	£34	£306.00
16	Yi Wei	3	122	£8	£976.00
17	Manuela Klein	1	22	£24	£528.00
18	Abu Kalkalakh	1	12	£36	£432.00
19	Sabina Lovrić	1	32	£28	£896.00
20	Xiu Juan Jen	3	99	£12	£1,188.00
21	Rajesh Singh	2	43	£18	£774.00
22	Elliot Bould	2	69	£24	£1,656.00
23	**TOTALS**				**£11,362.00**

4.5 Locating the Second Excel Document and Identifying Movie Titles

I performed a forensic investigation of the Local disk Drive A:\ drive to find out how many movie titles were mentioned in the second Excel file. I first discovered an Excel file called income.xlsx, but I also scanned for deleted files to make sure no other files were missed. One of the files I found is called **$RJKF6BP** in the **$RECYCLE.BIN** folder. It had an Excel icon, which meant it was another spreadsheet.

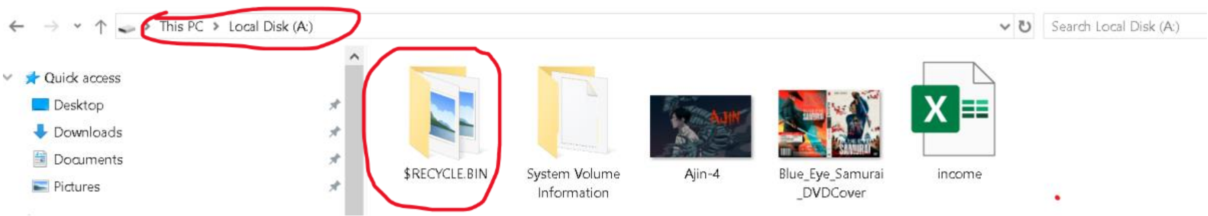
I opened **$RJKF6BP** and looked over the data to be sure it contained the correct information. After reviewing, I saw that there were eight (8) movie titles in the file. This verified that relevant information was contained in the second Excel file.

Actions Taken & Proof:

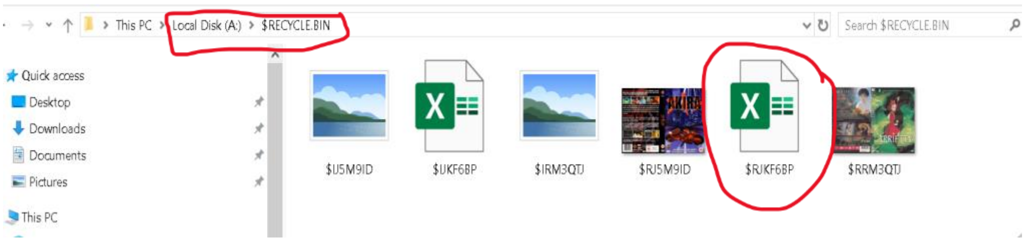
1. Located $RECYCLE.BIN by accessing the Local Disk
2. Discovered $RJKF6BP, which seemed to be an Excel document.
3. I opened the file and verified that it contained information on data of the movies
4. Confirmed eight (8) movie names and counted the total entries.

Screenshots as proof:

- **Screenshot 1:**



- **Screenshot 2:**



- **Screenshot 3, Final Answer:**

Movie Title	Territory	Main Contact	Sales 2017	Sales 2018
Charlie	North America	I.L. Licit	300000	378000
Rocks	North America	I.M. Moral	210000	311000
Crystal	North America	S.C. Andalous	119000	167000
Fire	Europe	C.R. Iminal	38000	62000
Horse	Europe	S.H. Ady	298000	345000
E	Asia	N.E. Farious	420000	465000
Mary J	North America	I.L. Legal	249000	288000
Speed	Europe	C. Rook	22000	120000
Totals			**1656000**	**2136000**

This shows how important it is to examine erased data in forensic investigations because important evidence can frequently be retrieved.

4.6 Evidence of potential crime in VeraCrypt Contents and in USB_03.E01

I discovered proof of a possible crime. Suspicious entries can be found in the Excel file **($RJKF6BP)** that was recovered from the **$RECYCLE.BIN.** It doesn't seem like the names under *"Main Contact"* are actual people. Rather, they appear to be coded allusions to illicit activity, implying an effort to hide transactions.

Furthermore, the high sales numbers might be a sign of fraud, money laundering, or other financial crimes. This file's presence in the Recycle Bin indicates that someone attempted to remove it to cover up evidence.

Listing of Concerning and suspicious names:

Listed Names:	Potential Meaning:
I.L Licit	Illicit
I.M Moral	Immoral
S.C. Andalous	Scandalous
C.R. Iminal	Criminal
S.H. Ady	Shady
N.E. Farious	Nefarious
I.L Legal	Illegal
C. Rook	Crook

Movie Title	Territory	Main Contact	Sales 2017	Sales 2018
Charlie	North America	I.L. Licit	300000	378000
Rocks	North America	I.M. Moral	210000	311000
Crystal	North America	S.C. Andalous	119000	167000
Fire	Europe	C.R. Iminal	38000	62000
Horse	Europe	S.H. Ady	298000	345000
E	Asia	N.E. Farious	420000	465000
Mary J	North America	I.L. Legal	249000	288000
Speed	Europe	C. Rook	22000	120000
Totals			**1656000**	**2136000**

Conclusion:

This forensic investigation used FTK Imager, Autopsy, HxD, and VeraCrypt to successfully identify encrypted, deleted, and concealed files. The results showed wiped out financial records, high-entropy files indicating encryption, and maybe illegal behaviour within encrypted files. Forensic analysis and legal proceedings depend on the accuracy and dependability of digital evidence. Decrypting more files and determining the full scope of the suspected crimes may necessitate more research.

References:

- OpenAI (2024) *ChatGPT (March 10 version)*. Available at: https://openai.com (Accessed: 10 March 2025).

- University of Roehampton (2025) *Lab1-FTKImager_Imaging Process_UG.pdf*. Available at: https://moodle.roehampton.ac.uk/mod/folder/view.php?id=1997482 (Accessed: 17 February 2025).

- University of Roehampton (2025) *Thumbdrive.E01* [Digital Forensic Image]. Available at: https://moodle.roehampton.ac.uk/mod/folder/view.php?id=1997482 (Accessed: 17 February 2025).

- University of Roehampton (2025) *Lab2 - An Introduction to Autopsy.pdf*. Available at: https://moodle.roehampton.ac.uk/mod/folder/view.php?id=1997483 (Accessed: 18 March 2025).

- University of Roehampton (2025) *Week3_Lab1_Hex_Editor.pdf*. Available at: https://moodle.roehampton.ac.uk/mod/folder/view.php?id=1997494 (Accessed: 18 February 2025).